8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

g) Quick Response Manufacturing (QRM)

Quick Response Manufacturing (QRM)

- Used to **increase operating efficiency** by the implementing a ***'flow production' system*** that is ***triggered by consumer demand***.

- ***Quick Response Manufacturing (QRM)*** is a company wide strategy to cut lead times in all phases of manufacturing and office operations. It can bring products to market more quickly and secure your business prospects by helping companies compete in a rapidly changing manufacturing arena.

Quick Response Manufacturing (QRM)

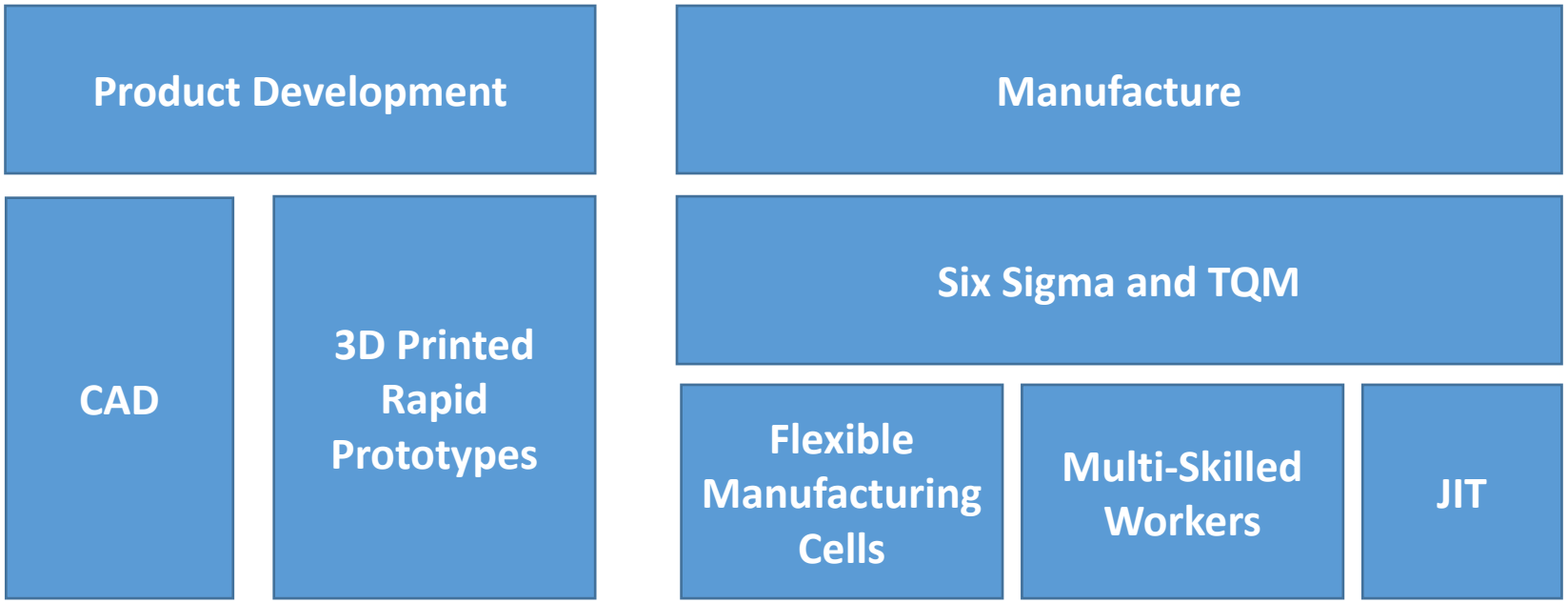
- ***Quick Response Manufacturing (QRM)*** requires the manufacturer to move from traditional 'batch' production to 'flow' production.
- ***QRM*** turns the company into one that responds to actual consumer demand rather than planning for an expected demand that may or may not happen.
- ***QRM*** involves several concepts – *Total Quality Management (TQM), Just in Time (JIT) and manufacturing cells*, but its main aim is to increase the *overall flexibility* and *responsiveness* of the company.

Quick Response Manufacturing (QRM)

- For example, by manufacturing in cells, production teams can be dedicated to specific product lines.
- These teams can be quickly and efficiently re-allocated if the requirements change.
- The manufacturer has increased production flexibility and will be better equipped to meet changing demands.
- No excess products are manufactured only those that are actually needed.

Quick Response Manufacturing (QRM)

Quick Response Manufacturing (QRM) is a ***'flow' manufacturing system*** for ***'low-volume, customisable production'***.



Key features of QRM

Quick Response Manufacturing (QRM)

- In an ideal QRM situation, the manufacturer would begin production as soon as an order is initiated.
- *Suppliers deliver raw materials directly to the production line, the product is manufactured, and the finished goods would flow directly to a waiting truck for delivery.*
- ***QRM*** is described as a ***'pull' process*** because the raw materials are ***'pulled'*** through the production process according to market demands.

Example of an organisation which employs ***QRM is the fashion retailer Zara***: https://www.tutorialspoint.com/case-study-of-the-growth-of-zara#:~:text=Fast%20Fashion%20%E2%88%92%20Zara%20is%20known,fashion%20retailers%20in%20the%20world.

Definitions:

Push System:

A ***push system*** relies on market demand and forecasting to dictate production levels. A pull system, however, uses current customer demand to dictate what to produce and when. ***Push systems*** are standard in industries that produce a wider variety of products and longer lead times.

Pull System:

Pull systems are usually seen in industries with shorter lead times. A ***pull system*** is a staple of a ***lean manufacturing model*** because it leads to ***lower inventory levels, less waste, and much faster response times to real-time customer demand***.

QRM example: designing and manufacturing a machine for packaging food

- Specification dictates a sequence of operations *(scoring, cutting, folding, filling, gluing)* customised to suit the packaging for a specific item of food.
- Design proposals are rapidly generated using CAD.
- CAD and physical prototypes are produced, including 3D printing.
- Transition into manufacturing is achieved as quickly as possible.
- Customers take delivery more quickly.

Advantages and Disadvantages of QRM for Manufacturers

Advantages	Disadvantages
Lower capital costs due to minimisation of raw materials and finished goods	Danger of being let down by suppliers and therefore being unable to fulfil orders
Larger market share likely due to the system's attractiveness to new clients	Problems resulting from wide fluctuations in demand
Demand-led small batch production, combined with quick turnaround, reduces stock and need for storage facilities	Potential human resource problems due to changing roles and responsibilities
Production teams accountability, combined with waste minimisation strategies, facilitate high quality at reduced costs	Requires multi-discipline support from all departments to facilitate successful implementation

Video: It's about time - this is QRM!!!

